[INCH-POUND]
A-A-59384
2 April 1999
SUPERSEDING
(See 7.5)

COMMERCIAL ITEM DESCRIPTION

PRESS, "H" FRAME, HYDRAULIC

The General Services Administration has authorized the use of this commercial item description in preference to MIL-P-80128, MIL-P-80149, A-A-55012, A-A-51200, A-A-55015 and A-A-51194.

1. SCOPE. This Commercial Item Description (CID) covers floor mounted "H" frame presses which are used in maintenance and repair shops to press, straighten, assemble, and disassemble parts.

2. CLASSIFICATION. The "H" frame press shall be of the following types, styles, classes and sizes. The type, style, class and size to be furnished will be as specified (see 7.1).

2.1 Type: Type refers to the type of bed provided on the press. The types described herein are as follows:

Type I - Non Roll Bed
Type II - Roll Bed (Stationary Frame)
Type III - Roll Frame (Stationary Bed)

2.2 Style: Style refers to the style of cylinder used on the press. The styles described herein are as follows:

Style 1 - Single action cylinder
Style 2 - Double action cylinder

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN.: DSCR-VBD, Richmond, VA 23297-5610.

FSC 3442

AMSC N/A
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2.3 Class: Class refers to the method of operation. The classes described herein are as follows:

Class A - Hand Operated
Class B - Electric Operated
Class C - Air Operated

2.4 Size: Size refers to the pressing capacity of the press. The sizes described herein are as follows:

010 - 10 ton capacity	040 - 40 ton capacity	080 - 80 ton capacity
017 - 17 ton capacity	050 - 50 ton capacity	100 - 100 ton capacity
020 - 20 ton capacity	055 - 55 ton capacity	150 - 150 ton capacity
025 - 25 ton capacity	060 - 60 ton capacity	200 - 200 ton capacity
030 - 30 ton capacity	075 - 75 ton capacity	300 - 300 ton capacity

3. SALIENT CHARACTERISTICS

3.1 Design. The press shall be new and of the "H" frame design. The press shall be designed for floor mounting and shall include all components, controls, parts, and features to meet the performance requirements specified herein. The press shall comply with the applicable requirements of Section 6 in ANSI B11.2, for the press ordered.

3.2 Controls. The press shall be equipped with all the controls required for the complete control of pressing operations. The controls shall provide for rapid advancement of the ram to, and retraction from, the workpiece. All operating controls shall be located convenient to the operator at his normal workstation.

3.3 Frame. The frame shall be fabricated of structural steel sections consisting of a bolster, work bed, siderails, and required support structures. All components shall be fastened together with bolts or by welding, or a combination of bolting and welding. The bolster shall support the hydraulic head assembly and shall provide the strength to maintain parallelism of the side members. The press frame shall have the strength to withstand not less than 150 percent of the maximum rated capacity of the press. The frame shall be supported by structures of the length and strength to provide stability to the press when loaded to maximum pressing capacity. The maximum and minimum ram nose to table dimensions shall be as specified (see 7.1).

3.3.1 Siderails and support pins. The siderails shall have a series of equally spaced holes to accommodate pins to support the work bed or bolster. Not less than two steel pins shaped for easy insertion and removal from the holes shall be provided. The shear strength of the pins shall be not less than 150 percent of the maximum rated capacity of the press. The pin hole spacing in all members shall be positioned to ensure that all pins support equal loads when pressing force is centered between the bolster and the work bed. Minimum width between the side rails shall be as specified (see 7.1).

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3.3.2 Type I: non roll bed. Type I non roll bed presses shall have beds with matching holes to align with the holes in the siderails for accommodating the support pins. The minimum width between bed channels shall be as specified (see 7.1). The bed shall have the strength to withstand the working forces when the press is operated at 150 percent of rated capacity. A hydraulic or mechanical means for raising and lowering the work bed shall be provided.

3.3.3 Type II: roll bed. Type II roll bed presses shall have beds mounted on rollers in tracks that allow the bed to roll out from under the bolster. The roll bed travel distance shall be such that when the bed is in the loading position, all bed edges shall be outside of a perpendicular line from the side of the bolster structure closest to the extended bed. The roll bed shall be provided with a locking mechanism to secure the bed in position. The bed shall have the strength to withstand the working forces when the press is operated at 150 percent of rated capacity. The minimum width between bed channels shall be as specified (see 7.1). When specified (see 7.1), a bed with adjustable channel width shall be supplied. A hydraulic or mechanical means for raising and lowering the bolster shall be provided.

3.3.4 Type III: roll frame. Type III roll frame presses shall be mounted on rollers that roll along the table axis in a track mounted on the side of the press table. The frame shall be provided with a locking mechanism to lock the frame in position. The press shall have the strength to withstand the working forces when operated at 150 percent of rated capacity. A hydraulic or mechanical means for raising and lowering the bolster shall be provided.

3.4 Hydraulic system. The hydraulic system shall conform to the requirements specified herein and National Fluid Power Association ANSI/(NFPA/JIC) T2.24.1. The safe operating pressure rating of all hydraulic components shall be 110 percent of the maximum rated capacity of the press. The hydraulic system shall consist of a hydraulic head assembly, fluid reservoir, pressure gauge and the necessary fittings and hoses to ensure a complete hydraulic system. The hydraulic system shall be either self-bleeding or have provisions for bleeding air from the system after the system is filled.

3.4.1 Hydraulic head assembly. The hydraulic head assembly shall support the cylinder and ram assembly. Unless otherwise specified (see 7.1), the hydraulic head assembly shall be permanently fixed to the center of the bolster. Moveable hydraulic head assemblies shall use roller bearings and shall be adjustable laterally along the bolster. A means shall be provided to lock the moveable hydraulic head assembly at any position within its range of travel. The moveable hydraulic head assembly shall be capable of performing full rated pressing operations at any position within its range of travel.

3.4.2 Cylinder and ram assembly. The cylinder and ram assembly shall be furnished with either a dual stage or single stage cylinder as specified (see 7.1). The minimum ram stroke shall be as specified (see 7.1). Provisions to prevent extension of the ram beyond its normal “end-of-stroke” position shall be provided. Hydraulic ram assemblies shall meet the requirements of ANSI B30.1. The ram axis shall be perpendicular to the worktable within an accuracy of 0.120 inch total indicator reading (TIR) per foot of travel.

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3.4.3 Fluid reservoir. The hydraulic fluid reservoir shall be capable of holding not less than 1-1/4 times the volume of fluid required to fill the hydraulic system and force the ram to the bottom of its stroke. If the reservoir is pressurized during any part of the operating cycle, the reservoir shall have a safe operating pressure rating of not less than twice the pressure that can be imposed on the interior of the reservoir.

3.4.4 Pressure gauge. A pressure gauge calibrated to read in tons of ram force shall be furnished. The upper range of the gauge shall be not less than 150 percent of the maximum system pressure. The pressure gauge shall comply with the requirements of ANSI B40.1, Grade B.

3.4.5 Pressure relief valve. The hydraulic system shall have a pressure relief valve, or similar device, to prevent loading the press beyond 110 percent of its rated capacity.

3.5 Work support devices. The press shall be furnished with two flat work support devices and two “V” type support devices. The devices shall be moveable along the entire length of the work bed. These devices shall have a means to prevent them from sliding off the work bed. A single pair of invertible support devices suitable for supporting flat work pieces on one side, and round work pieces on the other side, may be furnished in lieu of the two pairs. Regardless of the type of work support devices furnished, they shall withstand repeated pressing forces. One flat ram nose and one “V” ram nose shall be supplied.

3.6 Class A: hand operated hydraulic press. Hand operated presses shall conform to the following requirements.

3.6.1 Hydraulic pump. The hand operated hydraulic pump shall be capable of producing 10,000 psi when operated on shop air pressure of 80 to 100 psi. Maximum handle effort shall be less than 100 lbs.

3.7 Class B: electric motor operated hydraulic press. Electric motor operated hydraulic presses shall conform to the following requirements.

3.7.1. Power requirements. Electric operated presses shall be wired to conform to NFPA 79. Each press shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 7.1), each press shall be designed to operate on 3-phase, 60-Hertz (Hz) dual 230/460-volts. Conversion from one voltage to the other shall be accomplished by the reconnection of existing components.

3.7.2 Motors. The manufacturers standard motor provided to the commercial market shall be supplied.

3.8 Class C: air operated hydraulic press. Air operated hydraulic presses shall conform to the following requirements.

3.8.1 Air driven hydraulic pump. The power and performance characteristics of the air driven hydraulic pump and other components of the pneumatic system shall be sufficient to operate the

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press at 110 percent of maximum rated press capacity as specified herein, when operating on a shop air pressure of 80 to 100 pounds per square inch (psi). When specified (see 7.1), an air pressure regulator, pressure gauge, air filter, water separator, lubricator, and the necessary operating valves shall be provided. The pneumatic power system shall be mounted as an integral part of the hydraulic head assembly or as a separate unit mounted on the press frame. The pneumatic power system shall comply with the requirements specified herein and the requirements of ANSI B93.114M.

3.9 Standard equipment. All standard equipment normally provided with the manufacturers commercial press shall be furnished.

3.10 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described by the procuring activity (see 7.1).

3.11 Physical size limitations. When specified (see 7.1), the physical size of the press shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the press will fit into its future operating location.

3.12 Marking on instruments, control panels, charts, and plates. All words on the press, including all instruments, control panels, charts, plates and related attachments, shall be in the English language. Characters shall be marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.13 Data plates. All standard data plates normally provided with the manufacturers standard commercial press shall be furnished. Any additional data plates shall be as specified and fully described by the procuring activity (see 7.1).

3.13.1 Nameplate. Unless otherwise specified (see 7.1), a nameplate shall be securely attached to each press. The nameplate shall contain the information listed below.

Nomenclature
Manufacturer Name
Manufacturer Model
Manufacturer Serial Number
Contract Number or Order Number
National Stock Number
Date of Manufacture

3.13 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this commercial item description are met.

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4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE

5.1 Product conformance. The product provided shall meet the salient characteristics of this commercial item description; conform to the producer's own drawings, specifications, standards and quality assurance practices; and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

6. PACKAGING. Preservation, packing and marking shall be as specified in the contract or order (see 7.1).

7. NOTES

7.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in the order.

a. Title, number, and date of this specification.
b. Type (see 2.1).
c. Style (see 2.2).
d. Class (see 2.3).
e. Size (see 2.4).
f. Maximum and minimum distance ram nose to table (see 3.3).
g. Minimum width between side rails (see 3.3.1).
h. Minimum width between bed channels (see 3.3.2).
i. Adjustable bed channel width, if required (see 3.3.3).
j. Movable head assembly, if required (see 3.4.1).
k. Dual stage or single stage cylinder (see 3.4.2).
l. Minimum ram stroke length (see 3.4.2).
m. Electric power requirements, if different (see 3.6.1).
n. Air pressure regulator, pressure gauge, air filter, water separator, lubricator, valves, if required (see 3.7.1).
o. Optional equipment, if required (see3.10).
p. Physical size limitations, if required (see 3.11).
q. Additional data plate, if required (see 3.13).
r. Nameplate requirements, if different (see 3.13.1).
s. Packaging requirements (see 6).

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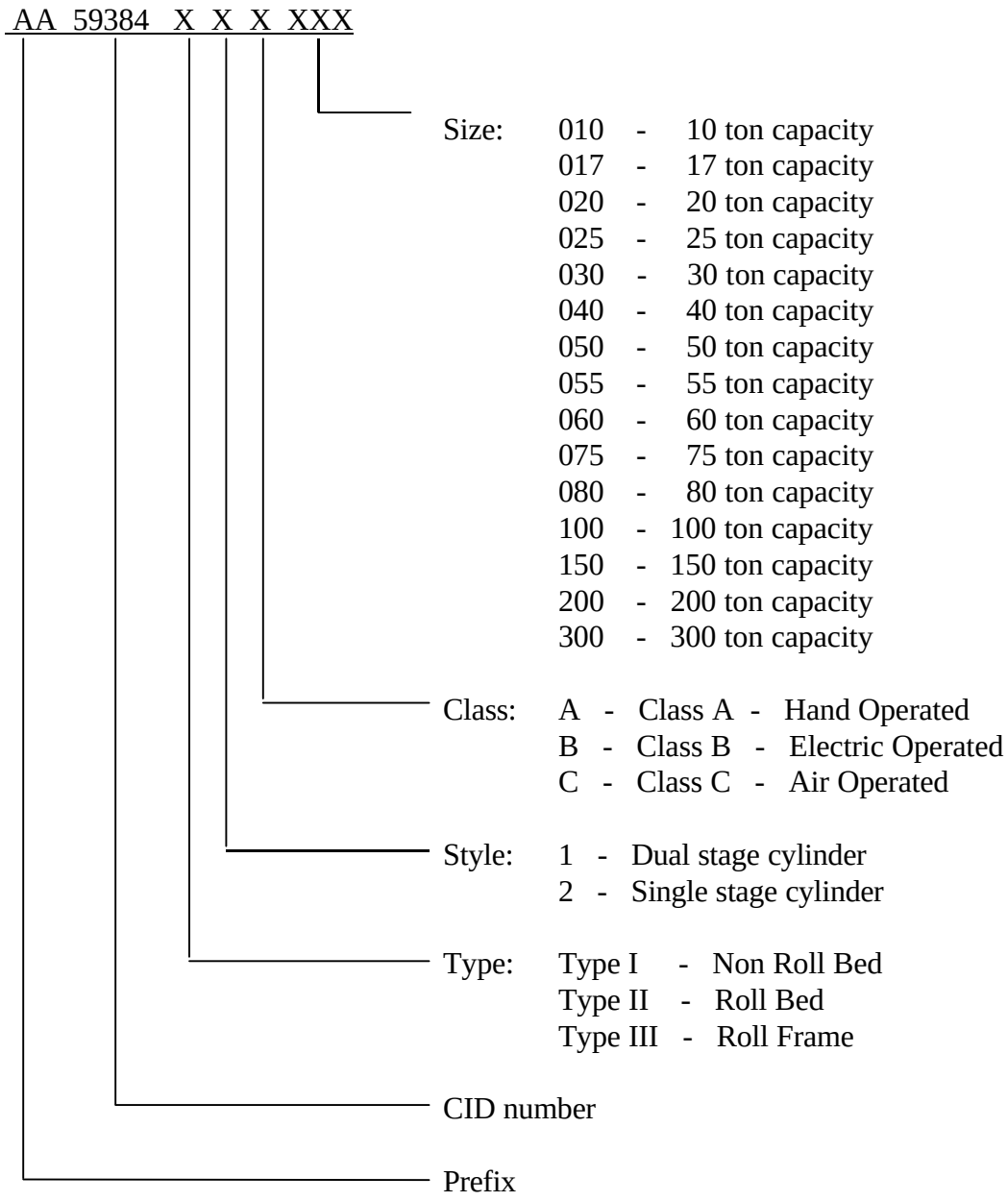
7.2 CID based part identification number. The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.



7.3 National stock numbers (NSNs). The following NSNs correspond to this CID. The list is not indicative of all possible NSNs associated with this document.

3442-01-368-6698	3442-01-206-5773	3442-01-384-7218
3442-01-382-0780	3442-01-382-0742	3442-01-381-9115
3442-01-382-0803	3442-01-382-5325	3442-01-382-1022
3442-01-382-1009	3442-00-471-9428	3442-01-382-0828

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7.4 Source for reference documents

7.4.1 ANSI standards T2.24.1, B40.1 and B93.114M are available from the American Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.

7.4.2 The Federal Acquisition Regulations (FAR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.4.3 NFPA 79 standard is available from the National Fire Prevention Association, Battery March Park, Quincy, MA 02269.

7.5 Supersession data. This CID supercedes MIL-P-80128, dated 29 January 1987, MIL-P-80149, dated 26 August 1985, A-A-55012, dated 28 April 1992, A-A-51200, dated 20 February 1991, A-A-55015, dated 28 April 1992, and A-A-51194, dated 2 February 1991.

MILITARY INTERESTS:

Custodians
Army - AL
Navy - SH
Air Force - 99

Reviewers
Navy - YD

CIVIL AGENCY COORDINATION ACTIVITY:
GSA - FSS

PREPARING ACTIVITY
DLA - GS

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